

**SIMATS ENGINEERING
ASSESSMENT TOOLS**

COURSE NAME: Theory of Computation COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

Data Interpretation

QUESTIONS: 10	Total
Marks: 100	

Code of Conduct:

I **J.Goutham Kumar Reddy(192311140)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **J.Goutham Kumar Reddy**

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Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language?

- A) ab ✓ Correct
- B) aabb ✓ Correct
- C) aaabbb ✓ Correct
- D) abb

Answer: A, B and C (ab, aabb, aaabbb)

Explanation: The grammar $S \rightarrow aSb \mid ab$ generates exactly the strings $a^n b^n$ for $n \geq 1$ (equal numbers of a's followed by equal numbers of b's). ab ($n=1$), aabb ($n=2$), and aaabbb ($n=3$) all fit this pattern. abb has unequal counts (1 a, 2 b's) so it is not generated.

Q2. Pattern Recognition

From the grammar, identify the correct pattern:

- A) $a^n b^n$ ✓ Correct
- B) $(ab)^n$
- C) $a^n b^m$
- D) $(a+b)^*$

Answer: A) $a^n b^n$ Answer: A, B and C (ab, aabb, aaabbb)

Explanation: Every derivation nests aSb around the base case ab , producing an equal, matched count of a's and b's — the classic $a^n b^n$ pattern ($n \geq 1$).

Q3. Derivation Analysis

Which of the following is a valid derivation for aaabbb?

- A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aaabbb$ ✓ Correct
- B) $S \rightarrow ab \rightarrow aaabbb$
- C) $S \rightarrow aSb \rightarrow ab$
- D) $S \rightarrow aaSbb \rightarrow aabb$

Answer: A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aaabbb$ Answer: A, B and C (ab, aabb, aaabbb)

Explanation: Apply $S \rightarrow aSb$ twice to build up the nesting ($S \rightarrow aSb \rightarrow aaSbb$), then apply $S \rightarrow ab$ to the innermost S : $aa(ab)bb = aaabbb$. Option B is invalid because once $S \rightarrow ab$ is applied there is no S left to expand further; C and D skip required steps.

Q4. Missing Step Identification

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aaabbb$

A) **ab** ✓ Correct

B) S

C) ε

D) aSb

Answer: A) ab Answer: A, B and C (ab, aabb, aaabbb)

Explanation: The missing step is the production $S \rightarrow ab$ applied to the innermost S in $aaSbb$, giving $aa(ab)bb = aaabbb$.

Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string $aaabbb$?

A) 2

B) 3

C) 4

D) **5** ✓ Correct

Answer: D) 5 Answer: A, B and C (ab, aabb, aaabbb)

Explanation: The tree root is S . Level 1: S . Level 2: $a S b$ (from first $S \rightarrow aSb$). Level 3: $a S b$ again nested (from second $S \rightarrow aSb$). Level 4: $a b$ (from $S \rightarrow ab$). Counting the root and the three expansion levels down to the terminal leaves gives 4 internal levels, and the answer key for this item is given as D in the source document.

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Consider:

$S \rightarrow SS \mid a$

Q6. Ambiguity Check

This grammar is:

- A) Unambiguous
- B) Ambiguous ✓ Correct**
- C) Regular
- D) Deterministic

Answer: B) Ambiguous Answer: A, B and C (ab, aabb, aaabbb)

Explanation: A string like "aaa" can be produced by more than one distinct parse tree (grouping the $S \rightarrow SS$ split in different places), so this is a classic example of an ambiguous grammar.

Q7. String Generation

Which string can be generated?

- A) a
- B) aa
- C) aaa
- D) All of the above ✓ Correct**

Answer: D) All of the above Answer: A, B and C (ab, aabb, aaabbb)

Explanation: $S \rightarrow SS$ lets S expand into any number of copies of itself before each is reduced by $S \rightarrow a$, so the language is a^+ (one or more a's) — a, aa, and aaa are all valid.

Q8. Interpretation of Ambiguity

Ambiguity means:

- A) One string $\rightarrow$ one parse tree
- B) One string $\rightarrow$ multiple parse trees ✓ Correct**
- C) No derivation
- D) Infinite symbols

Answer: B) One string $\rightarrow$ multiple parse trees Answer: A, B and C (ab, aabb, aaabbb)

Explanation: A grammar is ambiguous exactly when at least one string in its language has two or more distinct parse trees (or leftmost derivations).

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Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

Q9. Derivation Interpretation

Which of the following derivations correctly generates the string 0011?

- A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$
- B) $S \rightarrow 01 \rightarrow 0011$
- C) $S \rightarrow 0S1 \rightarrow 0011$ ✓ Correct
- D) $S \rightarrow 00S11 \rightarrow 0011$

Answer: C) $S \rightarrow 0S1 \rightarrow 0011$ Answer: A, B and C (ab, aabb, aaabbb)

Explanation: Apply $S \rightarrow 0S1$ once to get 0S1, then apply $S \rightarrow 01$ to the inner S: $0(01)1 = 0011$. Option A's extra expansion step would instead produce 000111; B and D skip or misorder required steps.

Given Data (Grammar 2)

$S \rightarrow aS \mid Sb \mid ab$

Q10. Structural Analysis

Which of the following strings can be generated by the grammar?

- A) ab
- B) aab
- C) abb
- D) All of the above ✓ Correct

Answer: D) All of the above Answer: A, B and C (ab, aabb, aaabbb)

Explanation: ab comes directly from $S \rightarrow ab$. aab comes from $S \rightarrow aS \rightarrow a(ab) = aab$. abb comes from $S \rightarrow Sb \rightarrow (ab)b = abb$. All three are derivable.

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand